

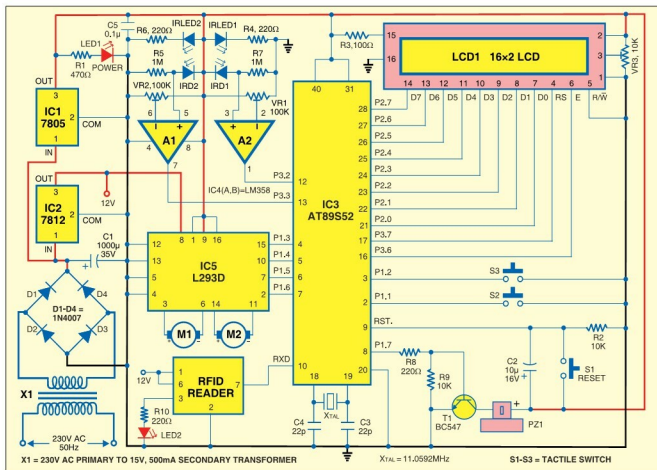


Fig. 5: Circuit of the automatic RFID-based automatic vehicle parking system

has to be parked, the RFID tag is placed near the RFID reader, which is installed near the entry gate of the parking lot. As soon as the RFID tag is read by the reader, the system automatically deducts the specified amount from the RFID tag and the entry gate boomer opens to allow the car inside the parking area. At the same time, the parking counter increments by one. Similarly, the door is opened at the exit gate and the parking counter decremented.

The system also offers the facility to recharge the amount for each RFID tag. No manual processing is involved. In addition, the system provides security.

Circuit description

Fig. 5 shows the circuit of the RFID-based automatic vehicle parking system. The circuit can be divided into different sections:

Power supply. Connector CON1

(refer Fig. 8), diodes D1 through D4, capacitor C1, and voltage regulator ICs 7805 (IC1) and 7812 (IC2) form the power supply section of the automatic vehicle parking system. CON1 is a three-pin connector that provides 15V AC or DC power supply to the circuit. In case of 15V AC, diodes D1 through D4 form a bridge rectifier to rectify the AC supply. Capacitor C1 filters out the ripples from the rectified output. ICs 7805 and 7812 provide regulated +5V and +12V, respectively, to the circuit. +5V is used to operate the microcontroller, LCD, RFID and IR sensor circuit and +12V operates the motor.

AT89S52 microcontroller. AT89S52 is a low-power, high-performance CMOS 8-bit microcontroller with 8Kb Flash memory. It is compatible with the industry-standard 80C51 instruction set and pin-out. The on-chip Flash allows the program memory to be reprogrammed

in-system or by a conventional non-volatile memory programmer. Other features include 256 bytes of RAM, 32 input/output lines, watchdog timer, two data pointers, three 16-bit timers/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry.

Connectors CON2 through CON4. CON2 and CON3 are two-pin connectors that connect the 12V DC motors to the circuit for controlling the entry and exit gate boomers. CON4 is a ten-pin dual-in-line female connector that connects the RFID reader module to the circuit.

L293D motor driver. H-bridge DC motor driver L293D (IC5) operates the DC motors to open the door or barrier for entry into and exit from the parking lot. Two high-current motor drivers can be used in place of L293D and 12V DC motors to control the entry and